Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports: A Synthetic Study

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Abstract. Operational flood streams contain duplicate, incomplete, and adversarial reports, while clustering and priority scores are often evaluated without measuring their scheduling consequences. We audit a fail-closed batch pipeline in which observable reports form a sparse graph, unresolved reports go to review, duplicate families feed a bounded score, and evaluator-only truth is joined after predicted-cluster scheduling. The clustering benchmark contains 80 synthetic runs from generator 3.0.0: additive Louvain has mean ARI 0.9191, product Leiden 0.9165, and product Louvain 0.9072. Product minus independently tuned additive is -0.0118 (bootstrap 95% CI $[-0.0226, -0.0024]$), but its Holm-adjusted test is not significant ($p = .2560$). A matched-density comparison reverses the point estimate but is also inconclusive. In a separate candidate-4.1 bundle, the revised score has NDCG@5 0.6669 versus 0.6763 for a random comparator and does not show a general alignment advantage; exact duplicates are invariant but coordinated high-confidence campaigns remain a failure mode. Dispatch results similarly show no general benefit over the legacy score and a clear cost relative to nearest-first. These are synthetic, within-generator diagnostics: boundedness and clustering accuracy do not establish source authenticity, policy validity, field benefit, or deployment readiness.

Keywords: Flood rescue · Graph clustering · Synthetic stress test · Priority ranking · Negative results.

1 Introduction and Related Work

During emergencies, crisis streams mix repeated reports of one event with nearby but distinct events, missing fields, and nonincident noise. Social-sensor and emergency-message research addresses volume, verification, and event extraction [9]. CrisisMMD and FloodNet capture useful disaster modalities, although

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FloodNet is an aerial post-flood scene-understanding dataset rather than a multimodal report stream [1,8]. These resources do not jointly provide physical-incident linkage, rescue priority, and field dispatch outcomes.

Social sensing and event extraction. Social-media users can act as real-time sensors [9], but the usual endpoint is event detection or classification rather than the consequences of sending a field resource to a predicted incident.

Consolidation and clustering. Graph community detection with Louvain or Leiden [3,13], and density methods such as DBSCAN and HDBSCAN [4] offer plausible consolidation mechanisms. Multiplicative spatial–range terms occur in bilateral filtering [12]; we adapt that structure to a product graph, while testing the exact gate and its bounds here.

Humanitarian logistics and dispatch. Relief-routing models expose competing equity, efficacy, and efficiency objectives [5]. We use this perspective to motivate a bounded priority ranking, but boundedness is not an expert-endorsed rescue policy. A key gap is propagation of false splits, merges, and noise into a scheduler that cannot consult incident truth.

We address these gaps with a fail-closed batch pipeline: reports with observable location and time enter a sparse graph, incomplete reports go to manual review, duplicate families feed a bounded ranking, and evaluator-only truth joins after predicted-cluster scheduling. We ask: (RQ1) does independently selected product composition improve clustering over additive composition? (RQ2) does the duplicate-aware score align with an independently generated benefit under report-level stress tests? (RQ3) how do predicted split, merge, and noise errors propagate into dispatch? The questions use two explicitly separated artifact bundles.

The contributions are (i) an observable-only pipeline with explicit review and evaluator boundaries; (ii) auditable threshold statements and a proposed duplicate-aware score; and (iii) a paired diagnostic reporting clustering, ranking, dispatch, and adverse results. We emphasize that the primary scientific value of this study lies not in a performance improvement but in the auditable negative and boundary results themselves: propagation errors and adversarial vulnerabilities that remain invisible to accuracy-only benchmarks are made explicit and quantified here. The evidence remains synthetic and within-generator, so the artifact is not a validated rescue policy.

2 Observable Pipeline and Methods

Figure 1 fixes the information boundary. Receipt-visible fields flow through graph construction, ranking, and a batch scheduler. Missing location or event time routes a report to review. Incident identity, benefit, deadline, and harm are stored separately and join only after a schedule exists.

Algorithm 1 restates the executable inference path as pseudocode: admission gating and manual-review routing (Section 2.1), similarity-graph construction

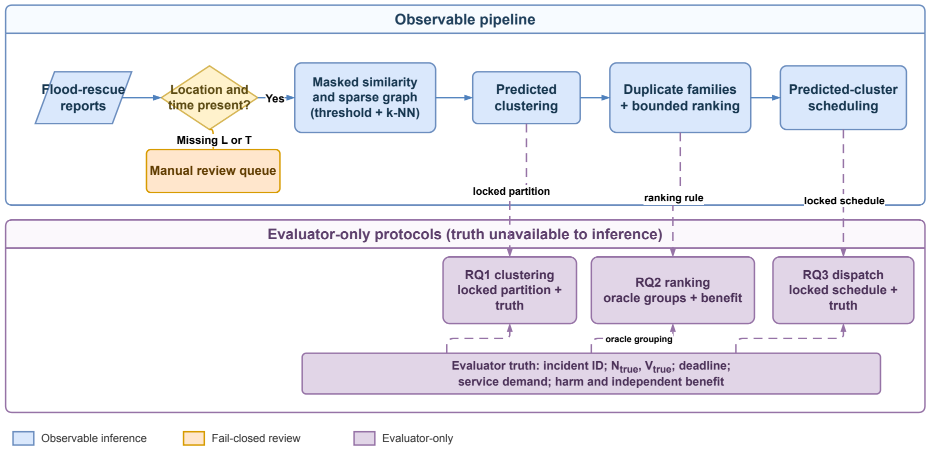

Fig. 1. Observable pipeline (solid), manual-review path (orange), and evaluator-only truth join (dashed). The scheduler receives predicted clusters, never incident truth.

and clustering (Section 2.2), duplicate-aware cluster scoring (Section 2.3, detailed as a subroutine in Algorithm 2), and scheduling. The evaluator-only truth join is described after the algorithm and is never an input to this inference path.

Require: Reports $R = \{r_i\}$; kernel scales $\sigma, \tau_t, \tau_F, \tau_E$; weights β, γ (and α for the additive composition); quantile q; neighbour cap k; Louvain/Leiden resolution ρ; composition $m \in \{\text{product}, \text{additive}\}$

Ensure: Dispatch schedule S; the algorithm below never reads incident identity, benefit, deadline, or harm labels

$R_{\text{obs}} \leftarrow \emptyset$; $R_{\text{review}} \leftarrow \emptyset$
for each report $r_i \in R$ **do**
$a_i \leftarrow \mathbf{1}\{L_i, T_i \text{ observed}\}$ ▷ Section 2.1
if $a_i = 0$ **then**
$R_{\text{review}} \leftarrow R_{\text{review}} \cup \{r_i\}$ ▷ orange manual-review path
else
$R_{\text{obs}} \leftarrow R_{\text{obs}} \cup \{r_i\}$
end if
end for
Compute each n_i^{corrob} from unique observable payloads in the full R_{obs}, then set
$Q_i \leftarrow \text{sigmoid}\big(b_0 + b_1 \mathbf{1}\{\text{image}_i\} + b_2 \log(1 + n_i^{\text{corrob}})\big)$ for every $r_i \in R_{\text{obs}}$
$E_0 \leftarrow \textsc{BuildSimilarityEdges}(R_{\text{obs}}, m, \sigma, \tau_t, \tau_F, \tau_E, \alpha, \beta, \gamma)$ ▷ BuildSimilarityEdges computes G, T, C on the full eligible pair matrix
$\theta \leftarrow q$-quantile of the nonzero weights in E_0
$E_\theta \leftarrow \{(i, j, w_{ij}) \in E_0 : w_{ij} > \theta\}$
$E \leftarrow E_\theta \cup$ (endpoint-wise top-k edges in E_θ), symmetrized by OR
$\mathcal{G} \leftarrow (R_{\text{obs}}, E)$
$\mathcal{C} \leftarrow \text{Louvain/Leiden}(\mathcal{G}, \rho)$ ▷ predicted clusters, never incident truth
for each predicted cluster $C_k \in \mathcal{C}$ **do**
$P_k \leftarrow \textsc{PriorityScore}(C_k)$ ▷ Algorithm 2
end for
$S \leftarrow \text{Scheduler}(\{(C_k, P_k) : C_k \in \mathcal{C}\})$ ▷ scheduler sees predicted clusters and P_k only
return S

Algorithm 1: Fail-closed observable report pipeline (Fig. 1)

After S is fixed, the offline evaluator may join S and R_{review} with incident identity, benefit, deadline, and harm labels to compute ARI, NDCG@5, and dispatch outcomes. This dashed path is not available to the inference algorithm. In the current artifacts, the core schema requires location and event time, so the manual-review branch is a fail-closed design boundary rather than an empirically exercised missingness condition.

2.1 Report Attributes and Missingness

Each report is represented by the compact tuple

$$r_i = (L_i, T_i, F_i, E_i, N_i, V_i, Q_i),$$

where L_i is a GPS location and T_i is the observable report timestamp ("created_at"). Flood evidence F_i and urgency evidence E_i are in $[0, 1]$; $N_i \in \mathbb{Z}_{\geq 0}$ is the reported trapped-person count; $V_i \geq 0$ is a vulnerability-evidence value; and $Q_i \in [0, 1]$ is a provenance/confidence heuristic. Q_i is not a calibrated probability of truth. The implementation also carries observable support fields such as image presence, source type, province, and missing-field flags; these support Q_i and duplicate fingerprints but are not additional latent labels.

The score Q_i is computed from observable image presence and the number of nearby corroborating observable payloads (counted by unique payload identity) using the proposed heuristic

$$\begin{aligned} Q_i &= \operatorname{sigmoid}\big(b_0 + b_1 \mathbf{1}\{\text{image}_i\} + b_2 \log(1 + n_i^{\text{corrob}})\big), \\ &(b_0, b_1, b_2) = (-.2, 1.4, .9). \end{aligned}$$

Here $\operatorname{sigmoid}(x) = (1 + e^{-x})^{-1}$. This is a pipeline rule, not a calibrated truth model. No independent receipt timestamp or source family is assumed.

Following the general missing-variable kernel problem [10], we use the following fail-closed construction. The admission indicator $a_i = \mathbf{1}\{L_i, T_i \text{ are observed}\}$. Thus $a_i a_j = 0$ prevents an edge involving a report whose geographic or temporal separation is undefined; such a report follows the manual-review path. Missing F_i or E_i is not zero-imputed: let $I_F(i, j), I_E(i, j)$ indicate that both reports share the corresponding observation and let $m_{ij} = I_F(i, j) + I_E(i, j)$; then

$$C_{ij} = \frac{m_{ij}}{2} \exp\left[-I_F(i, j)\frac{|F_i - F_j|}{\tau_F} - I_E(i, j)\frac{|E_i - E_j|}{\tau_E}\right], \quad C_{ij} = 0 \text{ if } m_{ij} = 0.$$

When $m_{ij} = 2$, both contextual fields are used; when $m_{ij} = 1$, the maximum contextual similarity is discounted to $1/2$; and when $m_{ij} = 0$, no contextual agreement is inferred. In the current 80-run artifact, every report has location, event time, flood, and urgency. Hence $a_i = 1$ and $m_{ij} = 2$ throughout the evaluated data, reducing the expression to

$$C_{ij} = \exp\left[-\frac{|F_i - F_j|}{\tau_F} - \frac{|E_i - E_j|}{\tau_E}\right].$$

The missingness extension is specified and unit-tested, but missingness robustness is not claimed as an empirical result of this benchmark.

2.2 Similarity Graph and Derived Bounds

For Haversine distance d_{ij} and time difference Δt_{ij}, define the Gaussian affinity [7] and exponential temporal decay [15]:

$$\begin{aligned} G_{ij} &= \exp[-d_{ij}^2/(2\sigma^2)], & T_{ij} &= \exp[-\Delta t_{ij}/\tau_t], \\ w_{ij}^{\times} &= a_i a_j G_{ij}(\beta T_{ij} + \gamma C_{ij}), & w_{ij}^{+} &= a_i a_j(\alpha G_{ij} + \beta T_{ij} + \gamma C_{ij}). \end{aligned} \tag{1}$$

The exponential form is a precedent, not a claim that the complete missingness-aware expression is copied from prior work [2]. The product is a spatial gate motivated by spatial–range products such as bilateral filtering [12]; both compositions and all weights here are defined by this paper. All scales $\sigma, \tau_t, \tau_F, \tau_E$ are strictly positive and the weights are nonnegative. The implementation forms the eligible pair matrix, computes an empirical nonzero-weight quantile, retains strict above-threshold edges, keeps the union of endpoint-wise top-k edges, and applies Louvain. Product and additive candidates have the same pool and grid, but each family is selected independently; the estimand is therefore a pipeline comparison, not an operator-only causal effect. Because product and additive pipelines are calibrated on separate quantile grids, reflecting how each would be tuned independently in practice, we treat this independently calibrated contrast as the primary estimand for RQ1 throughout the paper: it answers whether a practitioner who tunes each pipeline separately, as would occur in practice, obtains a net benefit from product composition. Section 4 also reports a matched-density contrast, which equalizes retained-edge fraction and mean degree across compositions, as a secondary robustness diagnostic that isolates the composition operator from calibration choices.

Theorem 1 (Product edge localization). *Let $B = \beta + \gamma > 0$. If a product edge with threshold $0 < \theta < B$ is retained, then $d_{ij} < \sigma\sqrt{2\log(B/\theta)}$. For $\theta \geq B$ the strict retained set is empty; for $\theta \leq 0$ the threshold alone gives no finite cutoff.*

Proof. Since $a_i a_j \leq 1$ and $T_{ij}, C_{ij} \leq 1$, $w_{ij}^{\times} \leq B\exp[-d_{ij}^2/(2\sigma^2)]$. Combining $\theta < w_{ij}^{\times}$ with positivity and taking logarithms gives the bound; the endpoint cases follow from the same inequality.

This is an edge statement. A component-diameter claim additionally needs a bounded observed hop diameter; long transitive chains remain possible.

Corollary 1 (Conditional component bound). *In the finite product region, a connected component with unweighted observed hop diameter $h < \infty$ and geographic diameter D satisfies $D < hr_\theta^{\times}$, where $r_\theta^{\times} = \sigma\sqrt{2\log(B/\theta)}$. A singleton has $h = D = 0$.*

The result follows from the triangle inequality along a path of at most h retained edges. It is a conditional post-hoc component bound, not an ex-ante compactness guarantee: the pipeline does not control h.

For the additive form, assume $\alpha > 0$. If $B < \theta < B + \alpha$, every retained edge satisfies

$$d_{ij} < r_\theta^+ = \sigma\sqrt{2\log\left(\frac{\alpha}{\theta - B}\right)}.$$

If $\theta \geq B + \alpha$, the strict retained set is empty; if $\theta \leq B$, no non-trivial threshold-induced geographic cutoff exists in general. Indeed, retention gives $\theta < \alpha G_{ij} + \beta T_{ij} + \gamma C_{ij} \leq \alpha G_{ij} + B$. Thus product composition has a finite geographic edge bound throughout its nonempty positive-threshold domain, whereas additive composition has such a bound only above the maximum non-geographic contribution. Neither statement alone implies better empirical clustering.

2.3 Duplicate-Aware Bounded Ranking

Exact duplicate payloads share a canonical fingerprint. Near duplicates are joined by deterministic connected components under the observable envelope; this transitive rule is not complete linkage. For a predicted cluster C_k, let $\mathcal{F}_k$ denote the resulting exact/near-duplicate evidence families inside that cluster. The revised estimator uses one confidence-gated priority per cluster and family maxima so that overlapping reports are not treated as disjoint people. Write $\widetilde{N}_i = \min(N_i, N_{\text{ref}})$ and $\widetilde{V}_i = \min(V_i, V_{\text{cap}})$ for the nonnegative clipped claims:

$$\bar{E}_k = \frac{\sum_{g\in\mathcal{F}_k} \max_{i\in g} Q_i E_i}{\max(1, \sum_{g\in\mathcal{F}_k} \max_{i\in g} Q_i)}, \qquad \bar{F}_k = \max_{g\in\mathcal{F}_k} \max_{i\in g} Q_i F_i,$$

$$\bar{N}_k = \min\left(1, \frac{\log(1 + \max_{g\in\mathcal{F}_k} \max_{i\in g} Q_i \widetilde{N}_i)}{\log(1 + N_{\text{ref}})}\right), \qquad \bar{V}_k = \max_{g\in\mathcal{F}_k} \max_{i\in g} Q_i \widetilde{V}_i,$$

$$P_k = \left[1 + (\mu - 1)\tanh(\bar{V}_k/s)\right](\omega_E \bar{E}_k + \omega_F \bar{F}_k + \omega_N \bar{N}_k). \tag{2}$$

The candidate source snapshot associated with the RQ2/RQ3 artifacts specifies $\omega = (.34, .33, .33)$, $\mu = 2$, $s = 10$, $N_{\text{ref}} = 500$, and $V_{\text{cap}} = 50$. The formula is a proposed policy heuristic: maxima limit duplicate multiplicity, the logarithm compresses heavy-tailed demand, clipping bounds individual evidence, and the hyperbolic tangent saturates vulnerability amplification. Reliability-aware truth discovery motivates using Q_i but does not derive these maxima [14]. Transformation, normalization, and weights require sensitivity and expert validation [11,5]; Section 4.1 reports a first such analysis. Comparators are a multiplicity-sensitive legacy score, urgency-only, population-only, a simple linear score, stable random, and—for dispatch—nearest-first.

function PRIORITYSCORE(cluster C_k)
$\mathcal{F}_k \leftarrow$ ExactFingerprintGroups(C_k) and NearDuplicateComponents(C_k)
$\bar{E}_k \leftarrow \dfrac{\sum_{h \in \mathcal{F}_k} \max_{i \in h} Q_i E_i}{\max(1, \sum_{h \in \mathcal{F}_k} \max_{i \in h} Q_i)}$ ▷ one representative per evidence family
$\bar{F}_k \leftarrow \max_{h \in \mathcal{F}_k} \max_{i \in h} Q_i F_i$
$\bar{N}_k \leftarrow \min\Big(1, \dfrac{\log\big(1 + \max_{h \in \mathcal{F}_k} \max_{i \in h} Q_i \min(N_i, N_{\text{ref}})\big)}{\log(1 + N_{\text{ref}})}\Big)$
$\bar{V}_k \leftarrow \max_{h \in \mathcal{F}_k} \max_{i \in h} Q_i \min(V_i, V_{\text{cap}})$
$P_k \leftarrow \big[1 + (\mu - 1) \tanh(\bar{V}_k / s)\big] \big(\omega_E \bar{E}_k + \omega_F \bar{F}_k + \omega_N \bar{N}_k\big)$
return P_k
end function

Algorithm 2: PRIORITYSCORE: duplicate-aware bounded priority (Eq. 2)

3 Experimental Design

Data contract and splits. The study uses two explicitly separated artifact suites. RQ1 uses the tracked generator-3.0 bundle, a semi-synthetic dataset with geographic anchors from the Copernicus EMSR848 Central Viet Nam activation; Copernicus EMS, OpenStreetMap, and WorldPop provide geographic or historical anchors, while incidents, reports, duplicates, coordinated attacks, vulnerability, and labels are synthetic. It contains 80 runs with 16 latent incidents and 26,288 reports (17,901 genuine, 3,587 duplicate, and 4,800 coordinated non-incident reports), with 288–370 reports per run. Runs 1–20 are development, 21–40 calibration, and 41–80 the test split held out from calibration. These are within-generator tests, not mechanism-shift OOD transfer.

RQ2 and RQ3 use a separately generated Candidate-4.1 bundle. This bundle is a fully synthetic regional scenario defined by the generator and is not claimed to be EMSR848-anchored. It contains 80 datasets, 16 incidents per dataset, 30,229 reports in total, and 339–424 reports per dataset; the development, calibration, and test seeds are 1000–1019, 2000–2019, and 3000–3039. Across its splits there are 1,554 fake reports and 2,560 labeled duplicates (1,280 exact and 1,280 near). Each base dataset contains one five-report coordinated high-confidence campaign and one report for each of four low-confidence attack types; RQ2 stress scenarios perturb these inputs. For both suites, inference reads only observable `algorithm_input.json`; ground truth is opened only after predictions are fixed.

Selection and comparators. The current notebook evaluates complete small grids on the 20 calibration runs and selects mean ARI, breaking ties by standard deviation and canonical configuration JSON. Product and additive use fixed $\tau_F = .25$, $\tau_E = .35$, and $\beta = \gamma = .5$. Selected product Louvain uses $\sigma = 700$ m, $\tau_t = 60$ min, quantile .90, $k = 8$, and resolution 1.2. Selected additive Louvain uses $\alpha = 1$, $\sigma = 700$ m, $\tau_t = 60$ min, quantile .95, $k = 8$, and resolution 1.2. Because selected quantiles differ, this is an independently calibrated pipeline comparison, not a matched-density operator comparison. Comparators are a convex similarity, product Leiden, geo-time DBSCAN, DBSCAN/HDBSCAN on scaled observable features, spatially constrained agglomerative clustering, and coordinate K-means. The implemented geo-time DBSCAN lacks ST-DBSCAN's non-spatial criterion and is named accordingly.

Endpoints and inference. ARI on incident-linked reports [6] is the primary endpoint; NMI, pairwise F1, geographic diameter, runtime, false destinations, fake absorption, noise rejection, retained-edge fraction, and mean degree are descriptive or operational diagnostics. The benchmark includes a matched- density product–additive comparison. Seed-paired ARI differences use 10,000 bootstrap resamples, Wilcoxon signed-rank tests, and Holm correction; RQ2 and RQ3 paired comparisons use 5,000 bootstrap resamples with the locked analysis seed. The reported p-values use explicitly exported analysis families. RQ1 applies one Holm family to all nine Product-Louvain-versus-comparator ARI contrasts; the three product/additive/Leiden contrasts are the primary comparisons and the remaining baselines are secondary diagnostics. RQ2 uses five comparator-specific three-endpoint families for alignment (NDCG@5, Spearman, and top-five overlap) and one five-endpoint family per robustness scenario (priority drift, mean-rank drift, top-k churn, N-error, and V-error); false priority lift is descriptive. For RQ3, each fixed candidate/comparator/context block applies Holm across all 14 dispatch endpoints. The primary RQ3 revision aggregates the three resource scenarios within each seed, so each paired test uses 40 seed pairs; the original 120 seed-by-scenario analysis is retained as a diagnostic artifact. RQ2 alignment uses oracle incident groups only to isolate ranking, whereas RQ3 evaluates predicted clusters. The priority stress tests and dispatch evaluation use independent benefit, deadline, service, access-delay, and harm outputs; no clustering metric is substituted for those outcomes.

4 Results

In summary, across the 40 held-out test runs neither graph-clustering composition shows a statistically robust advantage: additive Louvain attains the highest mean ARI (.9191), but its margin over product variants does not survive multiple-comparison correction under either calibration framing. For priority ranking, the proposed duplicate-aware score removes exact-duplicate inflation but does not show a general alignment advantage over simpler comparators, and shows larger priority/rank drift than the legacy score under a single realistic adversarial pattern—a coordinated high-confidence reporting campaign. Downstream, these behaviors propagate into the dispatch simulation: the revised policy shows no established improvement in harm or deadline outcomes over the legacy score, and performs significantly worse than a naive nearest-first baseline on both. The remainder of this section presents the supporting statistical detail.

Additive Louvain has the highest mean ARI. Product minus independently tuned additive is -0.01184 with CI $[-0.02257, -0.00240]$, with 9 wins, 19 ties, and 12 losses; raw $p = .0987$ and Holm $p = .2560$. Product minus matched-density additive is $+0.00989$ with CI $[-0.00100, 0.02082]$ and Holm $p = .2560$.

Table 1. RQ1 benchmark-test summary over 40 runs. Intervals are bootstrap 95% confidence intervals; diameter is descriptive.

Method	ARI [95% CI]	Pair F1	Mean diam. (km)
Additive Louvain	.9191 [.8995,.9381]	.9247	.3477
Product Leiden	.9165 [.8979,.9351]	.9223	.3547
Convex Louvain	.9135 [.8940,.9320]	.9195	.3547
Product Louvain	.9072 [.8864,.9271]	.9138	.3585
Matched-density Additive	.8973 [.8763,.9175]	.9046	.3743
Coordinate K-means	.8176 [.7976,.8384]	.8292	.2977
Spatial agglomerative	.6235 [.5964,.6494]	.6506	4.1648
HDBSCAN	.5205 [.4565,.5811]	.5787	1.2527
Geo-time DBSCAN	.4978 [.4589,.5366]	.5450	.6244
DBSCAN, all features	.4741 [.4273,.5201]	.5488	.9340

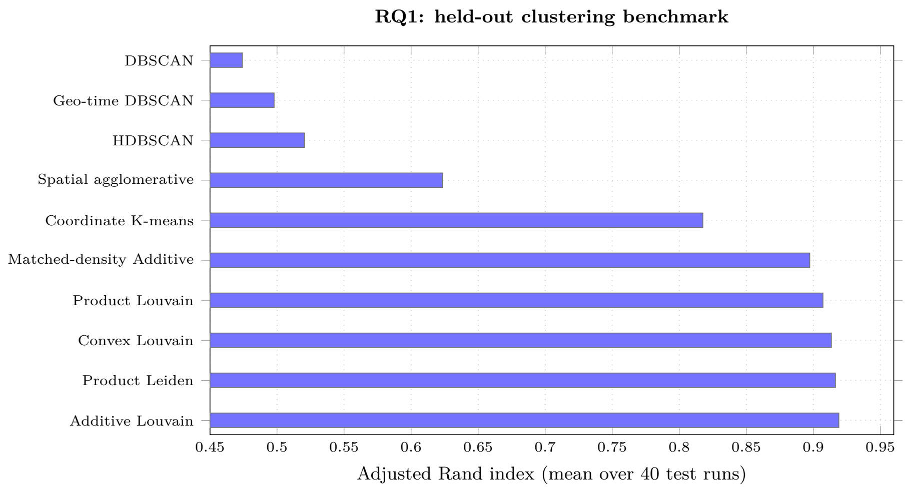

Fig. 2. RQ1 held-out ARI means generated from the benchmark summary CSV.

Product Leiden versus Product Louvain has Holm $p = .1120$. Thus neither composition advantage is established. Product and matched-density additive have nearly equal retained-edge fractions (.03011 and .03021) and mean degrees (9.90 and 9.94), so the matched-density contrast is a useful diagnostic but not a causal operator comparison.

The product pipeline has zero fake absorption and zero fake rejection but a 16.49% false-destination rate; fake reports avoid genuine destinations while still forming fake-only destinations. This is an adverse operational result, not evidence of misinformation robustness.

For RQ2, revised NDCG@5 is .6669 (CI [.6306,.7018]), Spearman is -.0304 (CI [-.1165,.0549]), and top-five overlap is .285 (CI [.2349,.3400]). The revised score is above legacy on NDCG@5 by .0121, but the Holm-adjusted contrast is not significant; it is below population-only (.6709) and random (.6763). Exact

Table 2. Headline RQ2/RQ3 results from the candidate-4.1 artifacts. RQ3 effects and p-values use 40 seed pairs after averaging the three resource scenarios within seed; the original 120 seed-by-scenario analysis is diagnostic. Positive paired effects favor the revised candidate; lower-is-better metrics use the artifact's oriented sign convention.

Question	Contrast and metric	Effect	Holm p
RQ2	Revised NDCG@5 (absolute)	.6669	–
	Random NDCG@5 (absolute)	.6763	–
	Revised − legacy NDCG@5	+.0121	.5882
	Exact-duplicate priority drift	0	–
	Coordinated-campaign drift (revised absolute)	.1688	–
	Revised − legacy campaign drift	−.1379	9.1×10^{-12}
RQ3	Revised − legacy, Product harm	−14.34	.563
	Revised − legacy, Product deadline	+.0188	.247
	Revised − nearest, Product harm	−69.88	4.1×10^{-5}
	Revised − nearest, Product deadline	−.1771	1.3×10^{-6}
	Product − matched Additive harm	−18.34	.000168

duplication at 2×, 5×, and 10× produces zero revised priority drift. A coordinated high-confidence campaign produces drift .1688 and false-priority lift .8828; revised drift is worse than legacy by −.1379 (Holm $p \approx 9.1\times10^{-12}$). Boundedness therefore does not guarantee ranking reliability.

For RQ3, after averaging the three resource scenarios within each seed, revised versus legacy on the Product partition has harm effect −14.34 (Holm $p = .563$) and deadline effect +.0188 (Holm $p = .247$), so no general improvement is established. Revised is significantly worse than nearest-first on both harm (Holm $p = 4.1 \times 10^{-5}$) and deadline (Holm $p = 1.3 \times 10^{-6}$). Product is also worse than matched-density Additive under the revised policy on harm (Holm $p = .000168$), and worse than oracle on the corresponding dispatch outcomes. Predicted split, merge, and fake-only destinations therefore propagate into operational consequences.

4.1 Sensitivity of Heuristic Constants

To assess robustness to the fixed constants in Q_i (Section 2.1) and P (Eq. 2), we define a local one-at-a-time sensitivity analysis on the Candidate-4.1 ranking task. Each valid constant is perturbed around its reference value while all other constants are held fixed, and NDCG@5 is recomputed on the same 40 test seeds. The ω variants perturb one component at a time and renormalize the vector; μ is restricted to its policy-valid interval $[1, 2]$. The locked rerun evaluates 21 configurations (one shared baseline and two or three tested levels for each constant) on the same 40 held-out seeds, yielding 840 seed-level observations. A constant is considered stable if the mean metric varies by less than 0.01 across its valid tested levels. Table 3 reports ranges copied from the sensitivity summary CSV; no value is inferred from the fixed- configuration comparisons.

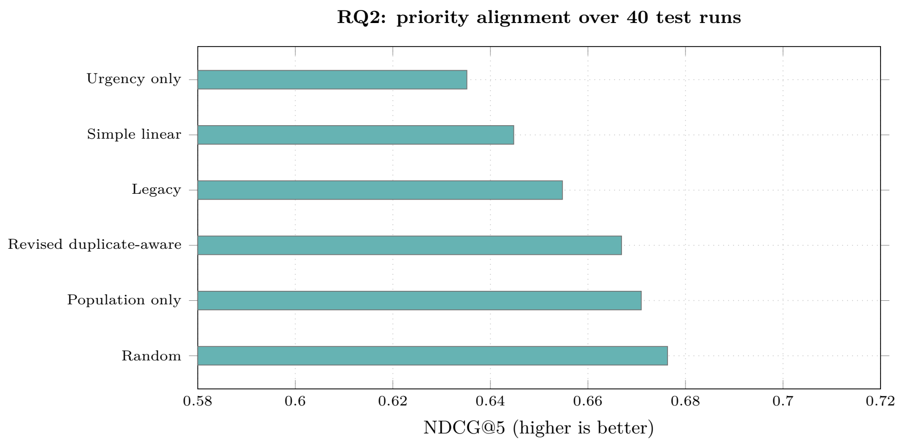

Fig. 3. RQ2 priority-alignment comparison generated from the candidate-4.1 summary CSV.

Table 3. Candidate-4.1 one-at-a-time sensitivity of mean NDCG@5 to fixed heuristic constants. Each range is the minimum–maximum mean over the tested levels, including the reference level, on 40 test seeds. All spans are below 0.01 under the locked rerun.

Constant	Reference/default value	Tested levels	Metric range
b_0 (Q_i intercept)	-0.20	$[-0.24, -0.16]$	$[.6653, .6672]$
b_1 (image weight)	1.40	$[1.12, 1.68]$	$[.6669, .6687]$
b_2 (corroboration weight)	0.90	$[0.72, 1.08]$	$[.6667, .6679]$
ω_E	.34	component $\times\{.8, 1, 1.2\}$, renormalized	$[.6663, .6714]$
ω_F	.33	component $\times\{.8, 1, 1.2\}$, renormalized	$[.6625, .6679]$
ω_N	.33	component $\times\{.8, 1, 1.2\}$, renormalized	$[.6663, .6697]$
μ (vulnerability amplification)	2	$\{1.6, 1.8, 2.0\}$	$[.6644, .6683]$
s (vulnerability saturation)	10	$[8, 12]$	$[.6665, .6713]$
N_{ref}	500	$[400, 600]$	$[.6669, .6669]$
V_{cap}	50	$[40, 60]$	$[.6669, .6669]$

5 Discussion and Threats to Validity

The results separate mathematical safeguards from empirical utility. Product gating supplies a finite edge bound in its positive-threshold domain, but the independently tuned product pipeline does not outperform additive clustering. The matched-density point estimate reverses the direction, yet its adjusted test remains inconclusive. The graph theorem is therefore a localization property, not evidence of a preferred composition.

The revised score removes exact-duplicate inflation and limits multiplicity, but it does not show a general alignment advantage. Its coordinated-campaign failure is especially important: an adversary can create a high-confidence pattern that remains numerically bounded while still perturbing the ranking. Dispatch results confirm that a bounded score cannot substitute for a validated opera-

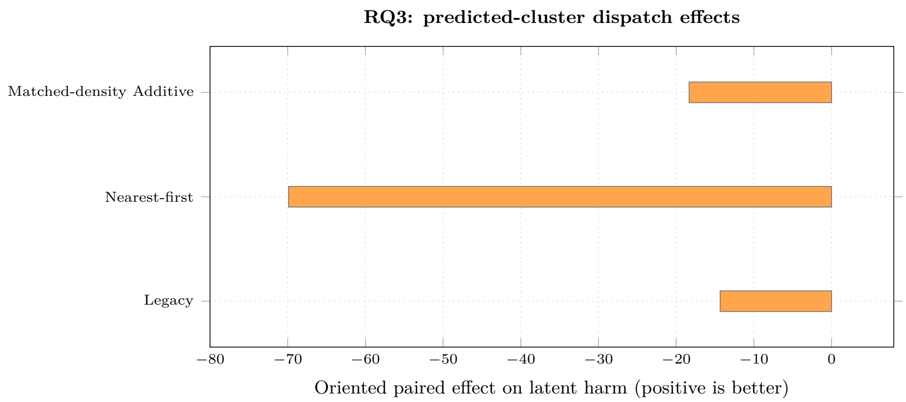

Fig. 4. RQ3 Product-partition paired effects on latent harm generated from the seed-level dispatch comparison CSV; positive values are improvement-oriented.

tional objective; clustering errors and fake-only destinations are transmitted to scheduling.

The local sensitivity analysis is stable over the tested intervals: every constant has a mean-NDCG@5 span below 0.01 across the 40 held-out seeds. The largest span is for ω_F (.00546), while changing V_{cap} from 40 to 60 leaves the mean unchanged at the reported precision. These results support numerical robustness within the tested neighborhood only; they do not validate the constants outside those levels or replace expert validation.

All reports, incidents, relevance signals, and dispatch outcomes are synthetic. RQ1 varies seeds within generator 3.0.0, while RQ2/RQ3 use a separate candidate-4.1 bundle; neither tests mechanism-shift OOD transfer or field performance. RQ1 does not provide an empirical missingness test because all evaluated reports contain the four core attributes. No expert panel validated the priority weights, caps, or utility. The study also does not measure extraction accuracy, device latency, memory, energy, provenance authenticity, or real-world harm. These limitations preclude policy or deployment claims.

6 Conclusion

The study establishes conditional graph bounds and documents competitive within-generator clustering, exact-duplicate invariance, and important negative results for ranking and dispatch. It does not establish ranking alignment, dispatch benefit, misinformation robustness, policy validity, or deployment readiness. The contribution is an auditable stress-test boundary around those claims, not a validated rescue policy.

Data and Code Availability

The accompanying artifact record contains the two data-suite manifests, benchmark notebooks, calibration grids, selected configurations, per-run test results, paired comparisons, the seed-level RQ3 reanalysis, and a deterministic figure-generation script. The publication figures are generated from result CSVs; every reported number is tied to an accepted CSV row and documented generator/configuration metadata. Raw upstream anchor snapshots are not redistributed in this artifact, and third-party materials remain governed by their own provenance and license records.

The analysis code and derived synthetic result tables are available at https://github.com/ngthtrong/nckh2. The rerun used generator commit `a6be3e9`; the RQ2 manifest records SHA-256 hashes for every reported CSV, including the 840-row sensitivity artifact and its 30-row summary.

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